

The role of extended reality for planning coronary artery bypass graft surgery

Madhurima Vardhan¹ *Harvey Shi¹ David Urick¹
Manesh Patel² †Jane A. Leopold³ ‡Amanda Randles¹

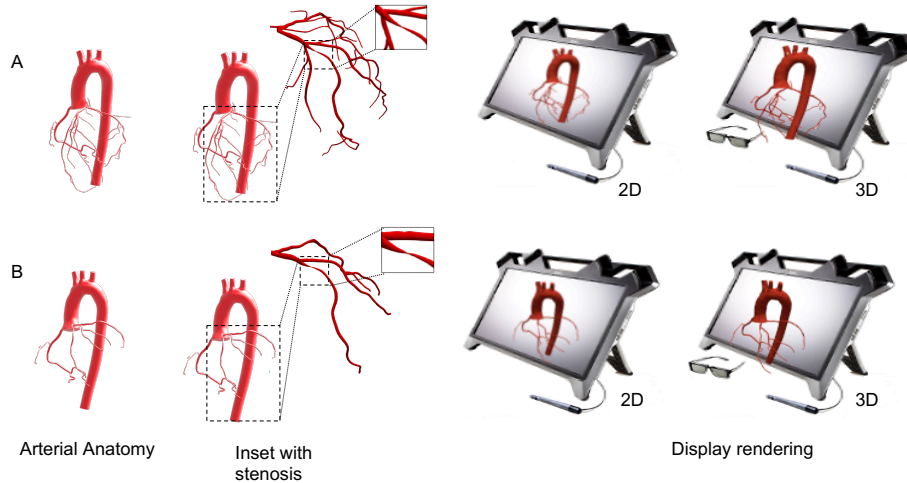


Figure 1: **Rendering arterial model on two types of display modes.** 3D/XR (with immersion) and 2D monoscopic (without immersion) displays both with stylus controller. A: High resolution models (all vessels $> 1mm$ diameter), B: Low resolution models (up to 1st and 2nd branching order). Insets identify the spatial region related to the location of coronary artery bypass graft placement.

ABSTRACT

Immersive visual displays are becoming more common in the diagnostic imaging and pre-procedural planning of complex cardiology revascularization surgeries. One such procedure is coronary artery bypass grafting (CABG) surgery, which is a gold standard treatment for patients with advanced coronary heart disease. Treatment planning of the CABG surgery can be aided by extended reality (XR) displays as they are known for offering advantageous visualization of spatially heterogeneous and complex tasks. Despite the benefits of XR, it remains unknown whether clinicians will benefit from higher visual immersion offered by XR. In order to assess the impact of increased immersion as well as the latent factor of geometrical complexity, a quantitative user evaluation ($n=14$) was performed with clinicians of advanced cardiology training simulating CABG placement on sixteen 3D arterial tree models derived from 6 patients two levels of anatomic complexity. These arterial models were rendered on 3D/XR and 2D display modes with the same tactile interaction input device. The findings of this study reveal that compared to a monoscopic 2D display, the greater visual immersion of 3D/XR does not significantly alter clinician accuracy in the task of bypass graft placement. Latent factors such as arterial complexity and clinical experience both influence the accuracy of graft placement. In addition, an anatomically less complex model

and greater clinical experience improve accuracy on both 3D/XR and 2D display modes. The findings of this study can help inform design guidelines of efficient and robust XR tools for pre-procedural planning of complex coronary revascularization surgeries such as CABG.

Index Terms: extended reality—coronary artery bypass graft surgery—anatomic complexity—treatment planning—stereoscopic and monoscopic displays

1 INTRODUCTION

New computational paradigms are emerging with rapid advancement in image processing to facilitate the entry of extended reality (XR) in pre-operative planning of surgical procedures [1, 17, 19]. Surgeons are usually early adopters of new technologies that can enhance operating room efficiency as surgical procedures can be challenging due to complex three-dimensional (3D) anatomic constructs [7, 18, 19, 22]. However, despite the continuous advances in the field of medical imaging, available imaging data in the form of X-ray images or computed tomography angiography (CTA) have been confined to two-dimensional (2D) screens [11]. Thus, the data-rich field of medical imaging is ripe for translating traditional 2D anatomic scans to immersive XR models. Such display technologies, due to greater immersion and interaction, warrant consideration for future applications in preoperative procedural planning to enhance surgical experience [7, 12, 23]. Therefore, this work proposes the use of XR technology in conjunction with conventional 2D display to view 3D arterial vasculature derived from 2D X-ray imaging data for planning coronary artery bypass graft (CABG) surgery.

CABG surgery is one of the gold standards for patients with advanced coronary heart disease, with over 400,000 cases performed each year in the US [15]. CABG surgery is an open heart surgery where an arterial or vein graft is used to bypass a significant block-

*¹Duke University. {mv82|harvey.shi|amanda.randles}@duke.edu

†² Duke University School of Medicine. {david.urick|manesh.patel}@duke.edu,

‡³Brigham and Women's Hospital. jleopold@bwh.harvard.edu,

age (stenosis) in a coronary artery [15]. Given the complexity and variability of the coronary arterial tree, it is crucial to understand anatomic features of coronary artery lesions within the context of the coronary artery anatomy to implant coronary bypass grafts successfully. Thus, there is a need for intuitive solutions to allow clinicians to interact and visualize arterial data in 3D for planning CABG surgeries.

Recent works have demonstrated the use of XR in pre-procedural planning of several cardiovascular surgeries [16, 19, 26]. XR has been shown to improve user performance in contexts including congenital heart disease, ventricular assist device fit testing, and visualizing distal coronary arteries in percutaneous revascularization procedures [4, 6, 10, 25]. However, other studies suggest that users performed better on 2D screens compared to XR for different types of tasks such as interacting with cardiovascular models, neurorehabilitation and 3D exploration [2, 3, 13, 21]. With conflicting findings from various studies, it remains unclear whether greater visual immersion from XR directly contributes towards operator accuracy and efficiency, questioning the need for the 3D/XR if the 2D interface suffices. Several studies comparing the effectiveness of the different 3D immersive and 2D displays also produce different conclusions [3, 13, 21, 25]. Moreover, in these studies, the ergonomics of tactile interaction also varied which further confounds the effect of visual immersion alone on the overall precision for performing a given task [3, 13, 21, 25].

The first goal of this study is to evaluate whether having greater visual immersion improves the accuracy of clinicians in a bypass graft placement task. Visual immersion depends on the stereoscopic display and can play a key role in understanding complex arterial anatomy. Two levels of visual immersion are evaluated: 1) with immersion (3D/XR) zSpace with the stereoscopic display on and 2) without immersion (2D) zSpace with the stereoscopic display off (Fig. 1). In the XR setup, use of the zSpace glasses in conjunction with the monitor enables stereoscopic viewing of the arterial models. The glasses also enable head motion parallax. In each, users can rotate and interact with the displayed objects with the zSpace stylus. The use of a unified software platform that supports both 2D and 3D/XR displays ensures that any deviations in CABG placement from a single user were due to differences in the visual immersion. Since the task of CABG placement can also be influenced by the peripheral device and the way this device is handled, the same input device is used with both 2D and 3D/XR displays.

The second goal of this study is to assess whether secondary factors could influence clinician accuracy. Latent variables such as the anatomic complexity of the arterial tree, and the experience level of the user could have an indirect influence on the efficacy of 3D/XR versus 2D for CABG planning. Within our study, we explored the effect of anatomic complexity by presenting users with two versions of each arterial tree: a high fidelity version representing the complex condition, and a simplified version representing low complexity. Finally, we also surveyed users on their background and training, allowing us to investigate whether experience level moderates task performance between the two visual conditions.

In this work we make the following contributions: we show that increased stereoscopic immersion did not affect user accuracy, but that user experience and the complexity of the visualized anatomy may be moderating factors. We believe these findings will help design future guidelines for XR tool development for pre-procedural planning.

2 METHODS

2.1 Patient-derived model reconstruction

Anonymized and de-identified imaging datasets were acquired from 6 patients who underwent both clinically indicated X-ray angiography and CTA within one month, under an approved Institutional Review Board protocol. A 3D reconstruction algorithm was used for

reconstructing 3D geometries from standard angiographic images separated by an angle $>45^\circ$ [5]. The 3D models were evaluated by two experienced cardiologists to confirm anatomic accuracy (vessel type and length). Two variations of a 3D arterial model having different anatomic complexity were created from the angiogram data: 1) high resolution: an arterial tree with all identifiable vessels and side branches up to 1 mm diameter, and 2) low resolution: a pruned version of the high resolution model that matched the vessel type, number, and length in the corresponding CTA data. Thus, 12 high and low complexity models were created from 6 patient datasets, as shown in Fig. 2. Arterial complexity was defined based on the number of side branches in the arterial tree [24]. To perform the bypass graft placement task, 95% stenoses were then introduced synthetically in the main vessels, right coronary artery, and left anterior descending for right (RCA) and left coronary arterial (LCA) trees, respectively, using open-source mesh software (insets in Fig. 2). The 12 arterial geometries were then rendered on 3D/XR and 2D displays.

2.2 Rendering arterial models on 2D and 3D/XR displays

Harvis software was used for rendering arterial geometries on 2D and 3D/XR display [21]. Harvis is a software platform built using the Unity Engine and is designed to be used for visualizing and interacting with arterial models on a variety of display modalities. In this study, the zSpace display (23.6-inch diagonal 1920 by 1080 pixel monitor) was used for rendering arterial geometries in 3D/XR through lightweight glasses. The zSpace monitor is used to project stereoscopic images of virtual arterial models into the real world space and allows for *in situ* interaction at the spatial position of the 3D hologram (Fig. 1). A stylus wand-style controller was used as the input device for users to interact with 3D arterial models. Users were able to examine the 3D models in detail from all angles and manipulate them using the stylus. The zSpace stereoscopic mode was turned off and was also used as a monoscopic 2D display (Fig. 1). In both the 3D/XR and 2D display modes, the input device was the stylus controller (Fig. 1). Keeping the input device consistent between both display modes prevented any potential deviations that could occur due to ergonomics of controller devices such as using a mouse on a 2D desktop. Both 2D and 3D/XR renderings were generated using Harvis. Therefore, using a unified software platform and same tactile interaction allowed us to determine the role of visual immersion for visualizing and interacting with 3D arterial models for CABG surgery.

2.3 Study description and quantitative user evaluation

In this work, 14 clinicians (9 males and 5 females) were recruited for the user evaluation based on power analysis using Wilcoxon signed-rank test ($\alpha=0.05$, effect size=0.8). To participate in the study, participants were required to have normal-to-corrected vision and advanced knowledge in the field of cardiology. Among the participants, 3 were attending physicians, with experience in the field of interventional cardiology of >10 years, and 11 were fellows-in-training in cardiology and interventional cardiology having experience in the field of cardiology of <5 years. Participants were not required to have any prior experience with 3D/XR and 71% of users had no prior experience, 22% used 3D/XR once or twice, and 7% used XR more than 3 times. All participants were required to complete a short tutorial to demonstrate the use of a 3D/XR display before starting the study and they completed the user study with both devices on the same day. No compensation was provided after the completion of the study.

Experimental task All clinicians were assigned the task of coronary bypass graft placement for 6 arterial geometries on both the 2D and 3D/XR display modes. Clinicians added a graft for each arterial geometry by specifying the preferred implant location by marking two dots for the proximal and distal location of the graft as

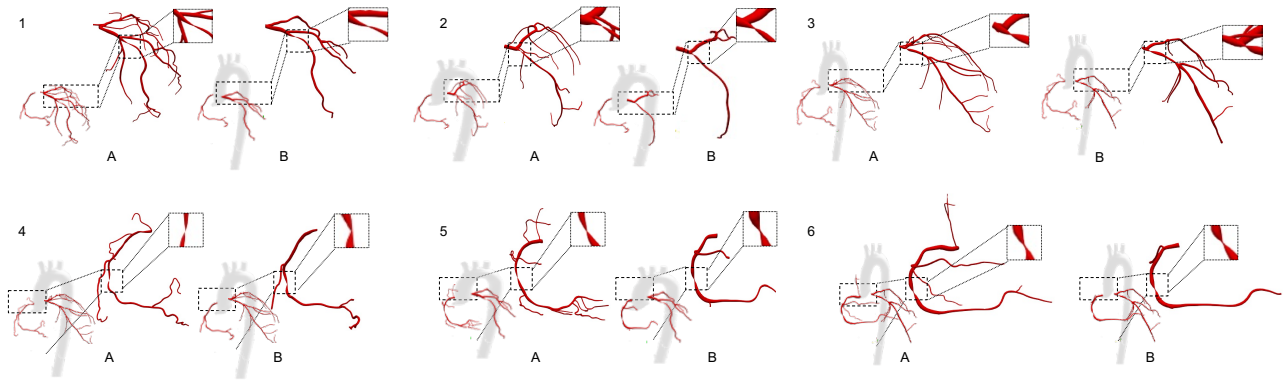


Figure 2: **Coronary artery geometries with two levels of anatomic complexity.** The insets identify the location of coronary artery bypass graft placement. 1-3 are left coronary arteries. 4-6 are right coronary arteries. A. High resolution models B. Low resolution models

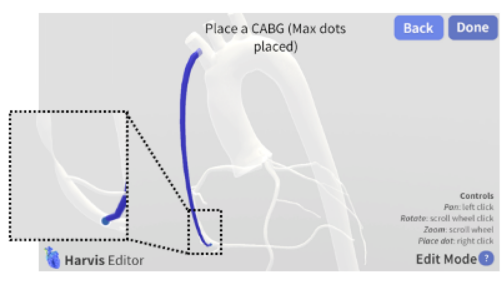


Figure 3: **Coronary artery bypass graft placement (CABG) using Harvis.** The graft placement between arterial vessels. Inset identifies the region related to the location of CABG placement.

shown in Fig. 3. Both high resolution and low resolution models were displayed to test the influence of anatomic complexity. Therefore, between the 6 arterial geometries on 2 display modes (2D and 3D/XR) and 14 participants, 168 total CABG placement trials were collected. The distal location of the graft was recorded for all trials. An inter-observer comparison between display modes was drawn for each unique clinician-task pair. At the end of the study, a post-survey was completed by participants to gather their thoughts on the utility and efficiency of XR in clinics and preference for using either 2D or 3D/XR displays.

Quantitative metrics and statistical analysis The correct bypass graft endpoints for all high resolution and low resolution arterial models were determined by an experienced interventional cardiologist who helped in the study design. These endpoints were treated as the ground truth and the distance deviation to the clinician-placed distal locations was computed. The lower the distance deviation from the ground truth bypass graft location, the higher the accuracy of the graft placement. We base our statistical procedures on fair practices for HCI, using confidence intervals and effect sizes [8]. The distance deviation metric was aggregated by the participant ($n=14$) before computing the confidence intervals. We computed the mean and the 95% bootstrapped pivot confidence interval and present the data in the following format: *mean*~[*CI lower bound*, *CI upper bound*].

Bias minimization Several measures were taken to minimize bias in the design of the study: 1) To ensure a consistent experience for all participants, the use of each display mode was preceded by an interactive tutorial where the clinician would learn all the navigational controls and how to place dots for creating the bypass

graft; 2) To prevent learning effects, randomized order of arterial geometries (both high resolution and low resolution) and display modes (2D and 3D/XR) was ensured for different participants; 3) To remove any potential effect from the peripheral pointer device, the same input device is used with both 2D and 3D/XR displays; 4) Experimental conditions were kept consistent, with the lighting and room setup being the same for all participants; 5) Instructions for module tasks were displayed on-screen, and quantitative metrics like distance deviation were recorded using the software itself; and 6) To control for arm fatigue, the session was designed to take between 10-15 minutes, and fatigue was addressed on the post-survey.

3 RESULTS AND DISCUSSION

3.1 Influence of visual immersion over CABG placement

The primary objective of this study was to assess the effect of visual immersion upon interactive visualization of arterial models for accurately determining the location of bypass graft insertion sites on coronary arteries as would occur in a CABG surgery. To isolate this effect, all clinicians were tasked to draw bypass grafts for 3 high resolution and 3 low resolution models on both display modes, 2D mode without immersion and 3D/XR with immersion, using the same stylus controller. Fig. 4 shows the overall accuracy results for 2D and 3D/VR display, quantified as the summed distance deviation between the clinician-placed CABG endpoints and ground truth data. Using the 2D display mode without immersion, distance deviation of $46.8 \sim [42.9, 50.9]$ mm was noted and on 3D/XR display mode with immersion $47.6 \sim [43.3, 51.7]$ mm was noted. Therefore, both display modes resulted in similar accuracy with 2D slightly better. The improved accuracy on 2D mode could potentially be because cardiologists during their training work extensively with 2D representations of imaging data (biplane angiogram). However, the overlapping CI suggest no significant differences in the overall accuracy on 2D and 3D/XR. Furthermore, all clinicians took similar time ($\sim 5-7$ minutes) for task completion using either display modes. These findings are also in line with the subjective responses of clinicians in the post-survey where 92% clinicians agreed that using either display mode would provide similar accuracy and efficiency.

Several studies have reported similar findings that 2D display outperformed XR with respect to precision and efficiency [2, 3, 13, 21]. However, few studies support the notion that XR improves accuracy for different tasks [9, 16, 25]. One such work suggested that XR immersion resulted in significantly higher accuracy for a stent placement task in arterial vessels [24]. However, this study used different input devices, a mouse and a stylus with 2D and 3D/XR, respectively, which could potentially result in tactile interactions to interfere with visual immersion. Another study that suggested that XR improved

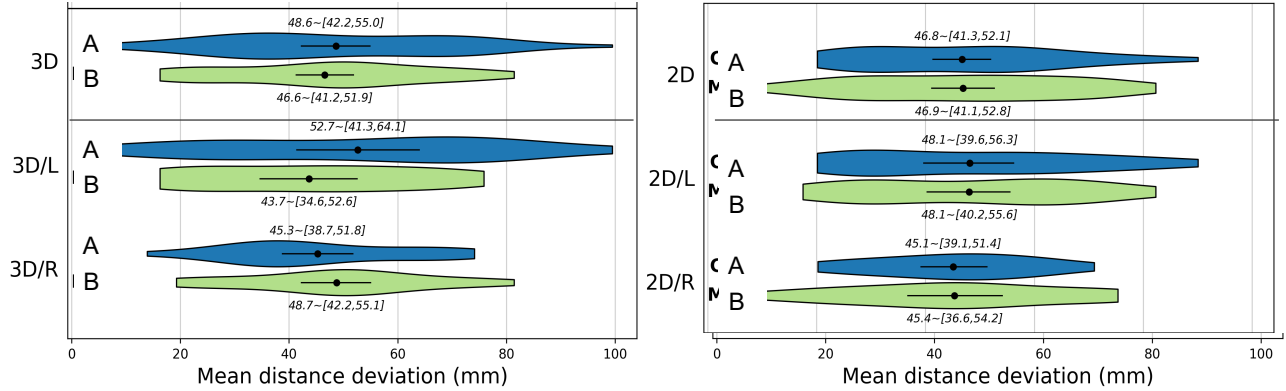


Figure 4: Overall mean distance deviation on 3D (left) and 2D (right) displays, broken down by: A. High resolution models, B. Low resolution models. Data=Mean~[lower and upper confidence interval]

physician accuracy in the case of spine fixation surgery compared XR to freehand surgery and not a 2D display [9]. Therefore, for planning CABG surgeries we conclude that both 2D and 3D/XR displays can provide similar overall accuracy suggesting that greater visual immersion may not necessarily improve performance.

Another cause for this discrepancy may be the types of tasks which are performed. While there are a multitude of studies reflecting the positive impact of VR, these findings cannot be taken all together. *McIntire et al* found that 60% of experiments rated VR positively in their 2014 review [14]. However, only 36% favored VR when limited to experiments regarding training and planning. The current finding further supports this notion that VR is not as useful for some tasks as it is for others.

3.2 Role of anatomic complexity in influencing bypass graft placement

In order to assess the role of anatomic complexity in influencing bypass placement, we first assessed the effect of capturing a larger number of side branches that would be included with higher resolution imaging enabling smaller vessels to be accurately resolved.

The second objective of this work was to assess factors besides visual immersion that may play a role in accurately placing the bypass grafts in arterial models. To this end, two factors were considered 1) anatomic complexity of the arterial model (high resolution models vs. low resolution models) and 2) left vs. right coronary arteries (right having fewer branches and inherently lower anatomical complexity). The effect of anatomic complexity was measured by breaking down the overall distance deviation for the high resolution and low resolution models (Fig. 4). For the high resolution models distance deviation was lower on 2D ($46.8 \sim [41.1, 52.1]$ mm) compared to 3D $48.6 \sim [42.2, 55.0]$ mm. Not only was the mean distance deviation lower on 2D, but the CI was also narrower suggesting lower variability on 2D compared to 3D/XR. This marginally improved accuracy on 2D could be attributed to the combined effect of familiarity and kinetic depth [3, 20]. On the other hand, for the low resolution models, the overall distance deviation between 2D and 3D was much closer with a similar distribution of CI ($46.9 \sim [41.2, 52.7]$ mm vs. $46.6 \sim [41.2, 51.9]$ mm).

To further understand the role of anatomic complexity, distance deviation for high resolution and low resolution arterial models was broken down for left and right arterial trees. The LCA tree is inherently more complex than the RCA due to a greater number of side branches and more secondary and tertiary branching patterns (Fig 2). It is noted that for LCA on 3D display (with immersion) a lower mean distance deviation with narrower CI was found for low resolution models compared to high resolution models

($43.7 \sim [34.6, 52.6]$ mm vs. $52.7 \sim [41.3, 64.1]$ mm). Mean distance deviation for RCA did not differ as much between low resolution and high resolution models on 3D display ($48.7 \sim [42.2, 55.1]$ mm vs. $45.3 \sim [38.7, 51.8]$ mm). However, on 2D display (without immersion) no difference in mean distance deviation for LCA between low resolution and high resolution models, although the CI was slightly narrower for low resolution models compared to high resolution models ($48.1 \sim [40.2, 55.6]$ mm vs. $48.1 \sim [39.6, 56.3]$ mm). Similar findings for the RCA were noted on the 2D display with negligible difference in mean distance deviation between low resolution and high resolution models ($45.4 \sim [36.6, 54.2]$ mm vs. $45.1 \sim [39.1, 51.4]$ mm). These findings suggest that depending on the spatial anatomic complexity, immersive 3D/XR displays potentially increase personal variability in objective performance influencing clinical decision-making. Visual immersion could potentially benefit clinicians differently and to varying extents based on the individual abilities of spatial understanding of complex anatomic features.

4 CONCLUSION

A quantitative user evaluation with clinicians was performed to assess the influence of visual immersion and latent factors such as anatomic complexity and clinical experience for planning CABG surgeries. Greater visual immersion offered by 3D/XR display did not significantly improve the accuracy for bypass graft placement compared to monoscopic 2D display. However, depending on the anatomic complexity greater immersion increased personal variability in objective performance measures relative to 2D displays. The findings of this study suggest that the intuitive notion of greater visual immersion resulting in improved accuracy may not always hold. Therefore, careful design consideration must be taken into account for content creation of treatment planning immersive tools. When comparing different immersive display modalities factors such as tactile interaction affecting ergonomics and device handling, anatomic complexity of the task, and experience of study participants should be appropriately examined. Furthermore, the conclusions reported in this study are generalizable to other controller-driven XR devices and users for a complex yet commonly performed surgical procedure. Moreover, our study population comprised of experienced clinicians and fellows-in-training who would generally be the common end-users of such technologies for pre-procedural surgical planning and training purposes.

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